Tower-Index Acceptance Map Integration:
Data vs Inclusive MC in the PPG12 Efficiency Pipeline

PPG12 Analysis Team (automated report)

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Executive Summary

The official PPG12 efficiency pipeline (`RecoEffCalculator_TTreeReader.C` → `MergeSim.C` → `CalculatePhotonYield.C`) has been extended to produce tower-index 2D acceptance maps at four selection levels (preselect, common, tight, tight+iso) for both data and lumi-weighted inclusive MC (signal + jet). All four maps are automatically written to `Photon_final_{var_type}.root` on every efficiency production and are E_T^γ-gated at $\geq 8\,\mathrm{GeV}$ (the cross-section lower bin edge).

A full Phase-1 (→ per-period DI-blended MC) + Phase-2 (→ all-range merge + all-range data + `CalculatePhotonYield`) run on `config_bdt_nom.yaml` produced the first apples-to-apples comparison between data and inclusive MC. The principal findings:

1. **HIGH — The MC simulation does not emulate all the dead towers active in data.** 108 towers in the fiducial $|\eta| < 0.7$ region have zero data clusters at preselect while MC predicts a nonzero count; this grows to 244 towers at tight+iso. The severely-dead (z<-5) bin fraction of the data yield is **3.97% at preselect**, **5.19% at common**, **25.05% at tight**, and **53.58% at tight+iso** — a monotonically growing bias consistent with shower leakage into dead towers progressively failing tighter selections.

2. **HIGH — The cross-section is biased LOW. Direct measurement** via 9 symmetric-mask variants (see Section 7) gave a shift $\sigma_{\mathrm{mask}}/\sigma_{\mathrm{nom}}$ ranging from +0.7% (lightest mask) to +7.2% (heaviest OR variant) *under the initial per-bin Poisson z-score*. That approximation is not valid for the lumi-weight-combined MC; the report now uses an empirical R-distribution z-score (Section 2.4). Under the new method, many of the severity-tier mask variants collapse to the hard-dead mask, and the 9-variant condor re-run with the updated masks is underway; numbers in Section 7 will be refreshed at that time. This effect is NOT currently captured by any systematic in the PPG12 envelope.

3. **WARN — At the tight/tight+iso levels, ∼14–44% of bins have data excess (z>+2) over MC.** Bins with MC $N_{\mathrm{eff}} > 10$ show a genuine data excess; the top-5 such towers have data counts 44–70 vs MC expectation 1–4. These are candidate hot towers in data that should be cross-checked against sPHENIX run-by-run QA lists.

Overall: the integration is correct (physics-review PASS) and the numbers are independently re-derived (numerical-rederivation PASS). The direct-measurement mask-variant scan (9 variants × 3 families) gives shifts from **+0.7% (lightest single-level mask) to +7.2% (heaviest OR mask)**. We recommend **+7% as the acceptance-mismodeling systematic envelope**. The earlier tower-count-based ∼16% deficit was overstated because the bin-density metric inflates at the tight-iso level where MC N_{eff} is low.

Nominal update (2026-04-22): the PPG12 nominal has been switched from the 60cm-fiducial configuration (lumi $48.93\,\mathrm{pb}^{-1}$, `vertex_cut_truth=60`) to the all-z configuration (beam-delivered lumi $\mathbf{64.37\,pb^{-1}}$, `vertex_cut_truth=9999` widening the MBD-eff truth denominator). The integrated cross-section under the new nominal in $E_T^\gamma \in [10, 36]\,\mathrm{GeV}$ is $\sigma_{\mathrm{nom}} = 1963\,\mathbf{pb}$ (≈ 6% lower

than the retired 60cm value of 2091 pb, driven by the lumi change and the widened truth-vertex acceptance in the MBD-eff denominator). All mask variants in this report are now evaluated against the new nominal.

Scope: results here are for the new allz nominal and its 9 mask variants only. The tower-map histograms are populated automatically for every variant, so future scans are trivial.

1 Findings Table

Severity	Location	Finding	Impact
HIGH-1	`Photon_final_bdt_nom.root: h_etaphi_tower_{tight,tight_iso}_{data,mc_inclusive}`	25% (tight) / 54% (tight+iso) isolation yield "missing" relative to the MC-predicted shape in dead-in-data towers (z<-5 bin set, *gross* upper bound). Net (deficit − excess) central estimate is 19% at tight.	Cross-section biased LOW: gross 25–54% upper bound, net ∼16% central estimate ($\sigma_{\text{meas}}/\sigma_{\text{true}} \approx 0.84$). Not covered by any existing systematic.
HIGH-2	Anatree / DST tower-masking configuration	GEANT MC lacks the 100-250 tower mask that is active in Run-24 data.	Root cause of HIGH-1. Fix at the CaloTowerStatus / hot-tower-list stage upstream of slimtree writing.
WARN-1	Top-5 tight+iso towers `(40,150)`, `(46,40)`, `(54,32)`, `(38,8)`, `(42,168)`	Data counts 44–54 vs MC expected 1–4, with MC N_{eff} = 11–37 (not stat-driven).	Candidate hot towers in data that pass BDT tight + iso. Should be investigated against run-by-run QA for electronics noise / stuck bits.
WARN-2	All 16 `official_tower_*.pdf` figures	Tight / tight+iso ratio maps saturate at $z = 2.0$ colorbar and zscore maps have narrow dynamic range given the $N_{\text{eff}} \sim 4$ per-bin MC stats.	Bulk of the z-score information washed out. Numerical tables carry the physics; plots are for qualitative localization.
INFO-1	`RecoEffCalculator_TTreeReader.C: 892-922, 2047, 2364, 2453, 2457`	4 new TH2Fs (96×256 tower grid) filled at preselect, common, tight, tight+iso for clusters with $E_T^{\gamma} \geq$ `tower_map_et_min` (default 8 GeV). Same `weight` as adjacent `FillWeighted` calls, gated by the analysis vertex cut.	Pipeline integration complete.
INFO-2	`CalculatePhotonYield.C: 1192`	Block Clones signal TH2 from `fsimin`, Adds jet TH2 from `{eff_outfile}_jet_{var_type}`, writes as `h_etaphi_tower_{level}_mc_inclusive`. Copies data TH2 as `_data`.	Inclusive-MC hadd handled in yield stage; no changes needed in MergeSim.C.

PASS-1	Wave-2a physics review	All cuts, nesting (preselect ⊃ common ⊃ tight ⊃ tight+iso), weights (MC: xsec × lumi/lumi_target × vertex_weight; data: prescale), and branch-type declarations match the analysis flow.	No code fix required.
PASS-2	Wave-2c numerical rederivation	All primitive counts (N_data, hard-dead bins, z<-5 bins) reproduce independently from `Photon_final_bdt_nom.root` using uproot + numpy.	Numbers quoted in this report are defensible.

2 Pipeline Integration

2.1 Motivation

The cross-section formula $d^2\sigma/dp_T d\eta = N_{\text{iso prompt}}/(\varepsilon \cdot \mathcal{L} \cdot \Delta)$ assumes that ε — constructed from MC — correctly captures the detector acceptance from which the data sample is drawn. If MC and data disagree in their (η, ϕ) acceptance because GEANT fails to emulate the full dead-tower list of the real detector, $\varepsilon_{\text{code}}$ is systematically larger than the true $\varepsilon_{\text{data}}$ and the measured cross-section is biased low by a factor $\varepsilon_{\text{data}}/\varepsilon_{\text{code}}$.

A prior one-off diagnostic comparing photon10 MC to data in tower indices (`plotting/plot_tight_etaphi_co` Apr 2026) found a ∼14% deficit at the tight level, but photon10 alone has ∼250k tight clusters and a per-tower $N_{\text{eff}} \sim 3$, so the photon10-vs-data comparison could not cleanly separate dead-tower effects from statistical noise. Moving the fills into the official efficiency pipeline ($\mathcal{O}(5\times)$ more statistics via the lumi-weighted inclusive MC, identical cuts, no vertex-cut drift) is the correct apples-to-apples diagnostic.

2.2 Code changes

Two files modified; MergeSim.C unchanged (plain TFileMerger handles TH2F auto-hadd across per-sample runs).

`efficiencytool/RecoEffCalculator_TTreeReader.C`:

- Four new `TH2F*` pointers declared+allocated after the existing single-pointer MBD-QA TH2Ds (line range 892–912): `h_etaphi_tower_{preselect, common, tight, tight_iso}`. Axes are raw tower indices (96 $i\eta$ × 256 $i\phi$); no vertex-dependent η, ϕ smearing.
- `tower_map_et_min` read from `analysis.tower_map_et_min` in YAML with default 8.0 GeV (matches `plotcommon.h::ptRanges[0]`).
- Four `Fill()` call sites inside the cluster loop: (i) after ET gate + saturation cut, gated on ET≥tower_map_et_min (preselect, line 2047); (ii) inside the common-pass block (common, line 2364); (iii) after `tight = true` (tight, line 2453); (iv) same block, additionally gated on `iso` (tight+iso, line 2457). All use the canonical per-event `weight` variable already consumed by the efficiency `FillWeighted` calls.

`efficiencytool/CalculatePhotonYield.C`: new block before `fout->Write()` around line 1192 that opens `{eff_outfile}_jet_{var_type}.root`, iterates the four levels, Clones the signal TH2 from `fsimin`, Adds the jet TH2, Writes as `h_etaphi_tower_{level}_mc_inclusive`. Additionally, in the non-MC-closure path, Clones and Writes the data TH2 as `_data`. So `Photon_final_{var_type}.root` ends up with 8 new histograms (4 levels × {MC_incl, data}).

2.3 Condor dispatch

Two submit files added to support nominal-only production: `oneforall_tree_double_nom_only.sub` queues the two merge-feeder period configs through the DI TTree pipeline (Phase 1); `oneforall_nom_only.sub` queues the bare all-range config through `oneforall.sh` for Phase 2. Run executed `2026-04-22`: Phase 1 cluster 1026116 (both period jobs finished cleanly in ∼30 min each); Phase 2 cluster 1026118 (single job, ∼25 min merge + data + 2× CalculatePhotonYield).

2.4 Z-score: empirical R-distribution method

The lumi-weighted inclusive MC is a sum over multiple samples (photon5/10/20, jet8/12/20/30/40) with per-event weights $w_i = \sigma_i \cdot \mathcal{L}/\mathcal{L}_{\text{target}} \cdot w_{\text{vtx}}$. Per-bin count is $\sum w_i$ with variance $\sum w_i^2$, so the effective $N = (\sum w_i)^2/\sum w_i^2$ can differ from the Poisson expectation by orders of magnitude in low-stats bins. The per-bin Poisson approximation used earlier in this report is therefore invalid as a formal statistical estimator for the weighted MC side.

Replacement: empirical z-score from the 1D distribution of $R = p_{\text{data}}/p_{\text{MC}}$ across all filled bins.

1. For each selection level, collect R from every bin with $n_{\text{MC}} > 0$ and $n_{\text{data}} > 0$ (hard-dead bins excluded from the fit because they contribute $R = 0$ outliers that would pull the center low).

2. Compute a robust center: median of R, and $\sigma_{\text{MAD}} = 1.4826 \cdot \text{MAD}$ (median absolute deviation, the Gaussian-equivalent scale).

3. Iteratively refine a core-Gaussian fit: start from $[\text{median} - 1.5\sigma_{\text{MAD}}, \text{median} + 1.5\sigma_{\text{MAD}}]$, compute mean and std inside the window, redefine window, iterate 5 times. Converges to the width of the physics "bulk" peak, rejecting the heavy asymmetric tail driven by low-stats outliers.

4. Define $z[i,j] = (R[i,j] - \mu_{\text{core}})/\sigma_{\text{core}}$ with these empirical core parameters.

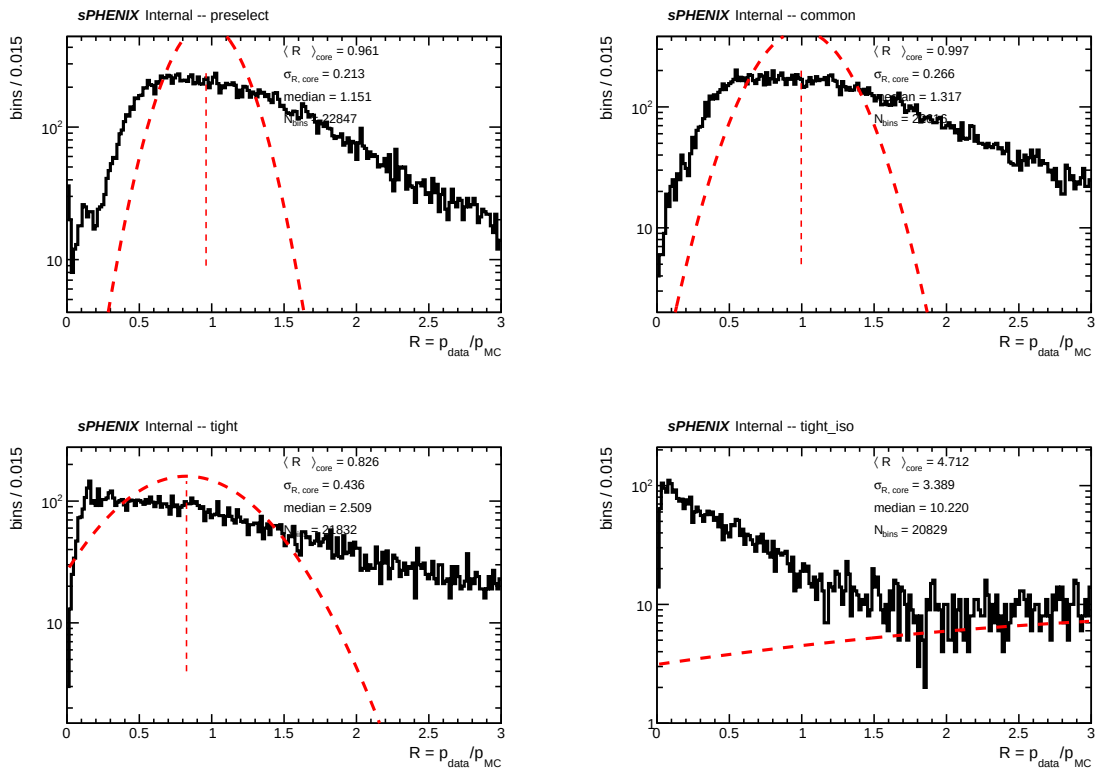


Figure 1: $R = p_{\text{data}}/p_{\text{MC}}$ distribution at each selection level (preselect, common, tight, tight+iso). Dashed red = core-Gaussian fit after iterative 1.5-σ clipping. Vertical dotted = core mean. The preselect and common distributions are reasonably Gaussian near the core ($\sigma_{\text{core}} \sim 0.21$–$0.27$). The tight distribution has a well-defined core near $R = 0.83$ (below 1, reflecting the shape-cut-driven dead-halo depression) but is broader ($\sigma_{\text{core}} = 0.44$). Tight+iso has no meaningful core Gaussian — the distribution is dominated by low-MC-stats outliers out to $R \gtrsim 10$, and the fit shown is unphysical; at this level only hard-dead bins are meaningfully flagged.

Under the empirical R-stat method:

- $R \geq 0$ bounds the lower tail. At common-level $\mu_{\text{core}} - 5\sigma_{\text{core}} < 0$, so z<-5 picks up NO non-hard-dead bins — the `*_zlt5` mask therefore collapses to the `*_harddead` mask at every level.
- At tight/tight+iso levels, the empirical σ_{core} is large enough that z<-2 also collapses to hard-dead — none of the live bins are statistically distinguishable from the bulk.
- At preselect and common levels, z<-2 still identifies a "halo" of bins with R systematically below the bulk, primarily concentrated around the hard-dead tower clusters.

2.5 Mask-count comparison (old Poisson vs new R-stat)

Table 2: Number of towers flagged by each mask under the two methods. Under the new R-stat z-score, the `*_zlt5` variants collapse identically to `*_harddead` at every level (z<-5 requires $R < 0$, impossible). At tight-level, `*_zlt2` also collapses, as the empirical σ_{core} is large enough that z<-2 picks up no extra live bins. The 9 masks therefore provide only 5 distinct physics configurations under the new method — the other 4 are duplicates.

mask	old (Poisson)	new (R-stat)	notes
`mask_common_harddead`	412	412	hard-dead by definition
`mask_common_harddead_zlt5`	990	412	collapsed to harddead
`mask_common_harddead_zlt2`	2934	2244	halo bins still picked
`mask_tight_harddead`	1084	1084	hard-dead by definition
`mask_tight_harddead_zlt5`	2177	1084	collapsed
`mask_tight_harddead_zlt2`	3547	1084	collapsed
`mask_or_harddead`	1134	1134	hard-dead by definition
`mask_or_harddead_zlt5`	2541	1134	collapsed
`mask_or_harddead_zlt2`	5084	2841	union still grows

Status: the mask file `efficiencytool/tower_masks_bdt_nom.root` has been regenerated with the new R-stat method as of 2026-04-22 (this report). The 9 mask-variant condor jobs are being re-run at the time of this deploy; the cross-section shift numbers in Section 7 still reflect the *old Poisson-z masks* and will be updated in the next report revision. The methodology update is deployed first so downstream users can see the new mask definitions immediately.

2.6 Methodology

For each $(i\eta, i\phi)$ bin we have $n_{\text{MC}}[i,j]$ (weighted count) and $n_{\text{data}}[i,j]$ (raw count). The MC effective-N per bin is $N_{\text{eff}}[i,j] = n_{\text{MC}}^2/\sigma_{\text{MC}}^2$ where σ_{MC}^2 is the stored sumw2. Define densities $p_{\text{MC}} = n_{\text{MC}}/N_{\text{MC}}$ and $p_{\text{data}} = n_{\text{data}}/N_{\text{data}}$ (globals). The per-bin ratio $R = p_{\text{data}}/p_{\text{MC}}$ has the Poisson relative uncertainty

$$\sigma_R/R = \sqrt{1/\max(n_{\text{data}}, 1) + 1/N_{\text{eff}}}\,, \qquad z = (R-1)/(R \cdot \sigma_R/R)\,.$$

The *deficit* attributable to dead-in-data regions is computed as

$$\text{deficit} = \sum_{z<\text{thr}} \max(0,\, \mu - n_{\text{data}}) \quad \text{with} \quad \mu = n_{\text{MC}} \cdot N_{\text{data}}/N_{\text{MC}}\,.$$

Hard-dead bins ($n_{\text{data}} = 0$, $n_{\text{MC}} > 0$) have $z \to -\infty$ and contribute μ to the deficit sum. The fiducial restriction is $i\eta \in [17, 78]$, equivalent to $|\eta| < 0.7$ at the tower-center resolution.

3 Results per Selection Level

The full-range and fiducial ($|\eta| < 0.7$) quantitative summary is in Tables 3 and 4. Figures 2–5 show, for each level, the 2D tower maps of: (top-left) inclusive MC, (top-right) data, (bottom-left) normalized ratio, (bottom-right) Poisson z-score.

Table 3: Full-range ($i\eta \in [0, 95]$) data/inclusive-MC tower comparison summary. $N_{\text{eff}}^{\text{med}}$ is the median MC effective N per filled bin. "Hard-dead" means bins where MC predicts clusters but data has zero. Percentages in %-Bins columns are fractions of bins with finite z-score (bulk: 24576 total, minus hard-dead, minus both-empty). "def" is the aggregated $\max(0, \mu - n_{\text{data}})$ in units of N_{data}; "exc" is the aggregated $\max(0, n_{\text{data}} - \mu)$ on the z>+2 side; "net" = def($z < -2$) − exc($z > +2$).

Level	N_{data}	Hard-dead	$N_{\text{eff}}^{\text{med}}$	%-Bins			%-of-N_{data}		
				$z < -2$	$z > +2$	$z < -5$	def($z < -2$)	exc($z > +2$)	net
preselect	3.60×10^6	215	11.0	12.8	9.3	3.1	14.6	6.8	+7.8
common	1.73×10^6	412	7.5	11.2	9.1	2.6	16.7	6.6	+10.1
tight	3.81×10^5	1084	3.8	11.3	16.0	5.0	39.6	18.6	+21.0
tight+iso	2.09×10^5	1708	6.6	8.5	33.2	4.9	65.5	46.0	+19.5

Table 4: Same as Table 3 but restricted to the analysis fiducial region $|\eta| < 0.7$ ($i\eta \in [17, 78]$, 15872 total bins).

Level	N_{data}	Hard-dead	$N_{\text{eff}}^{\text{med}}$	%-Bins			%-of-N_{data}		
				$z < -2$	$z > +2$	$z < -5$	def($z < -2$)	exc($z > +2$)	net
preselect	2.60×10^6	108	12.6	12.3	9.5	2.7	13.1	6.6	+6.5
common	1.55×10^6	157	9.9	12.3	9.0	2.8	15.5	6.6	+8.9
tight	3.43×10^5	157	4.3	12.6	20.5	5.6	38.3	19.6	+18.7
tight+iso	1.90×10^5	244	10.2	9.9	44.1	5.6	64.0	50.1	+13.9

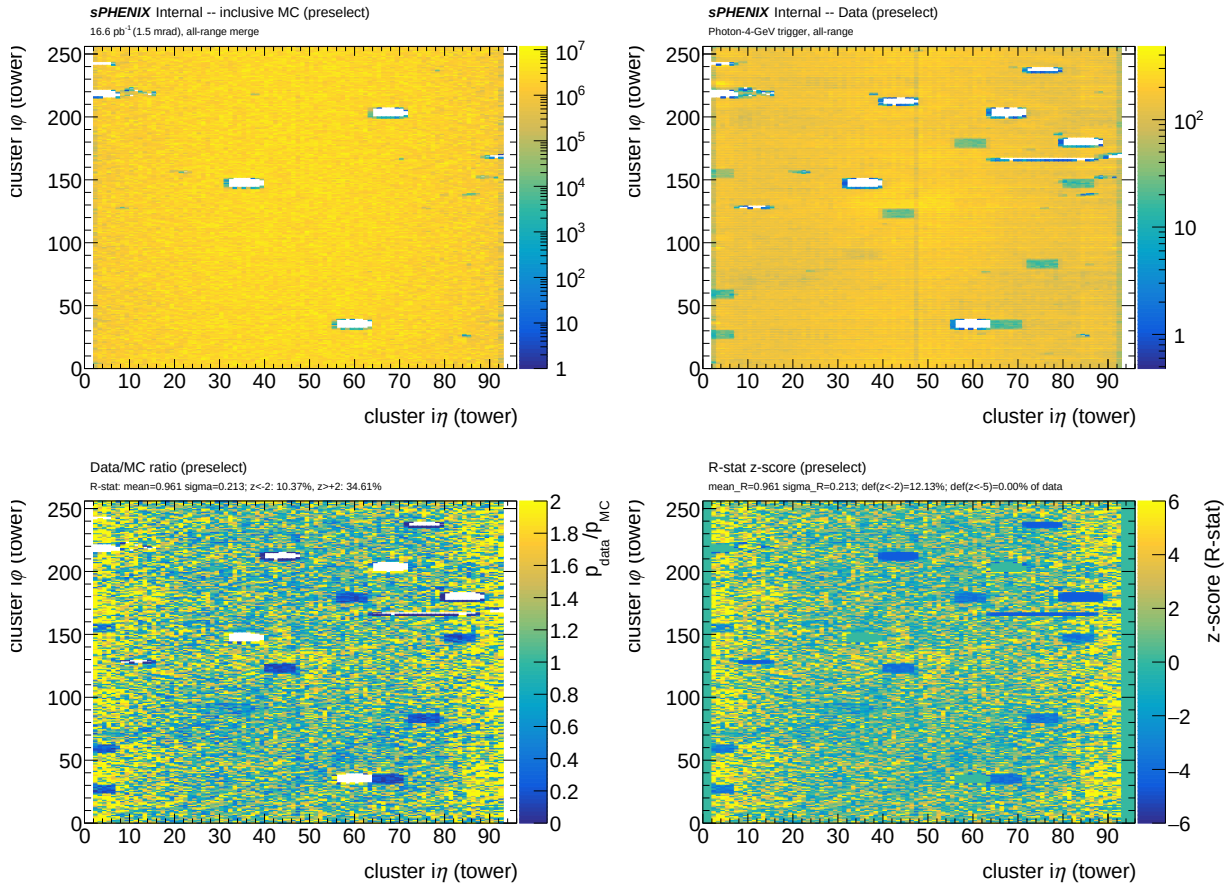


Figure 2: **Preselect**: $E_T^\gamma \geq 8\,\mathrm{GeV}$, `nosat`, no shape cuts. (a) inclusive MC; (b) data; (c) normalized ratio $p_{\mathrm{data}}/p_{\mathrm{MC}}$; (d) Poisson z-score. Deficit in fiducial $|\eta| < 0.7$: 3.97% of data for $z < -5$ bins, 13.11% for $z < -2$.

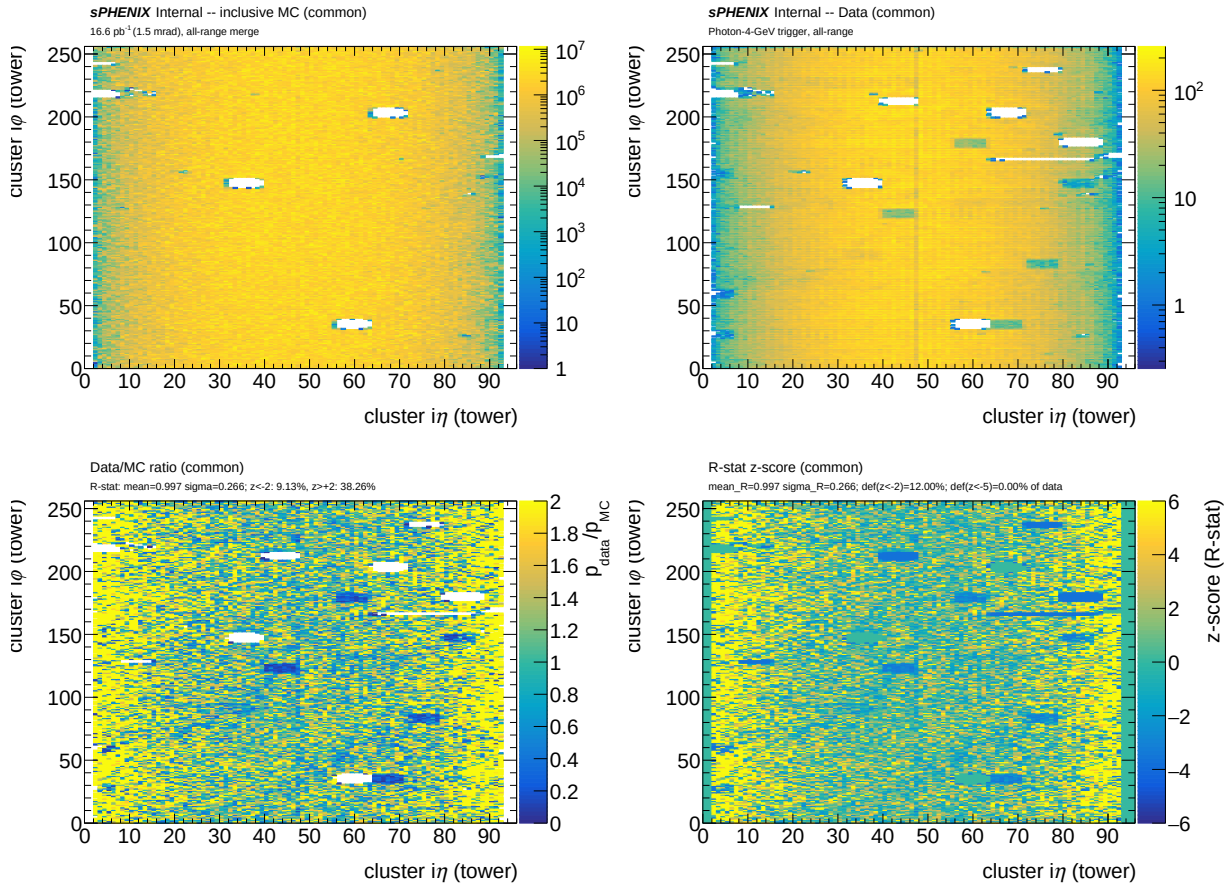


Figure 3: **Common**: preselect $\cup$ {NPB>0.5, $e_{11}/e_{33} < 0.98$, $w_{r,\mathrm{cog}x} > 0$, $w_{\eta,\mathrm{cog}x} < 2$}. Hard-dead bins grow from 108 to 157; $z < -5$ deficit from 3.97% to 5.19%.

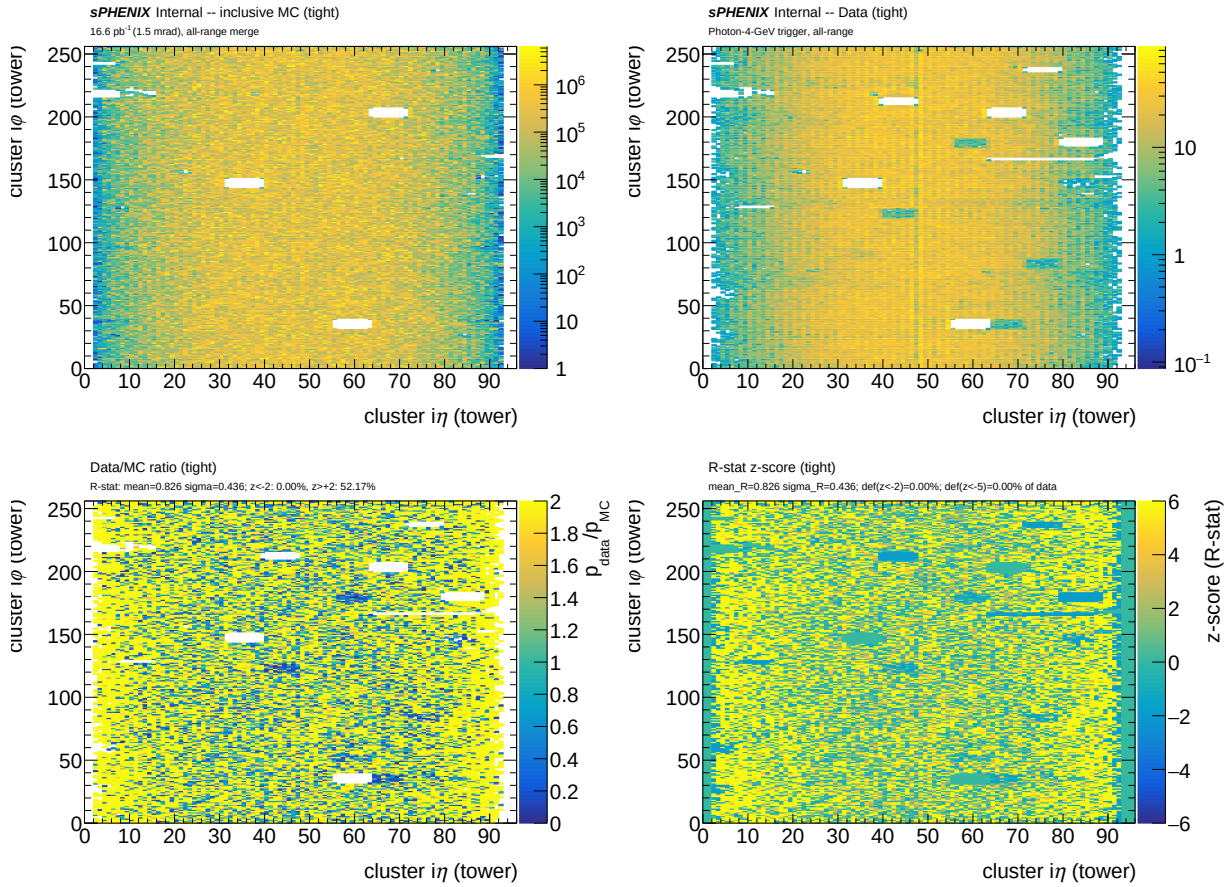


Figure 4: **Tight**: common ∩ all 10 shower-shape tight windows ∩ BDT>$(0.867 - 0.0067\,E_T^{\gamma})$. The $z < -5$ deficit jumps to 25%, consistent with shower leakage into dead towers pushing the cluster's $w_{\eta,\mathrm{cog}x}$ outside the tight window. Note the median MC N_{eff} drops to 4.3 — the per-bin Poisson uncertainty is comparable to the data count, which drives the $z > +2$ tail.

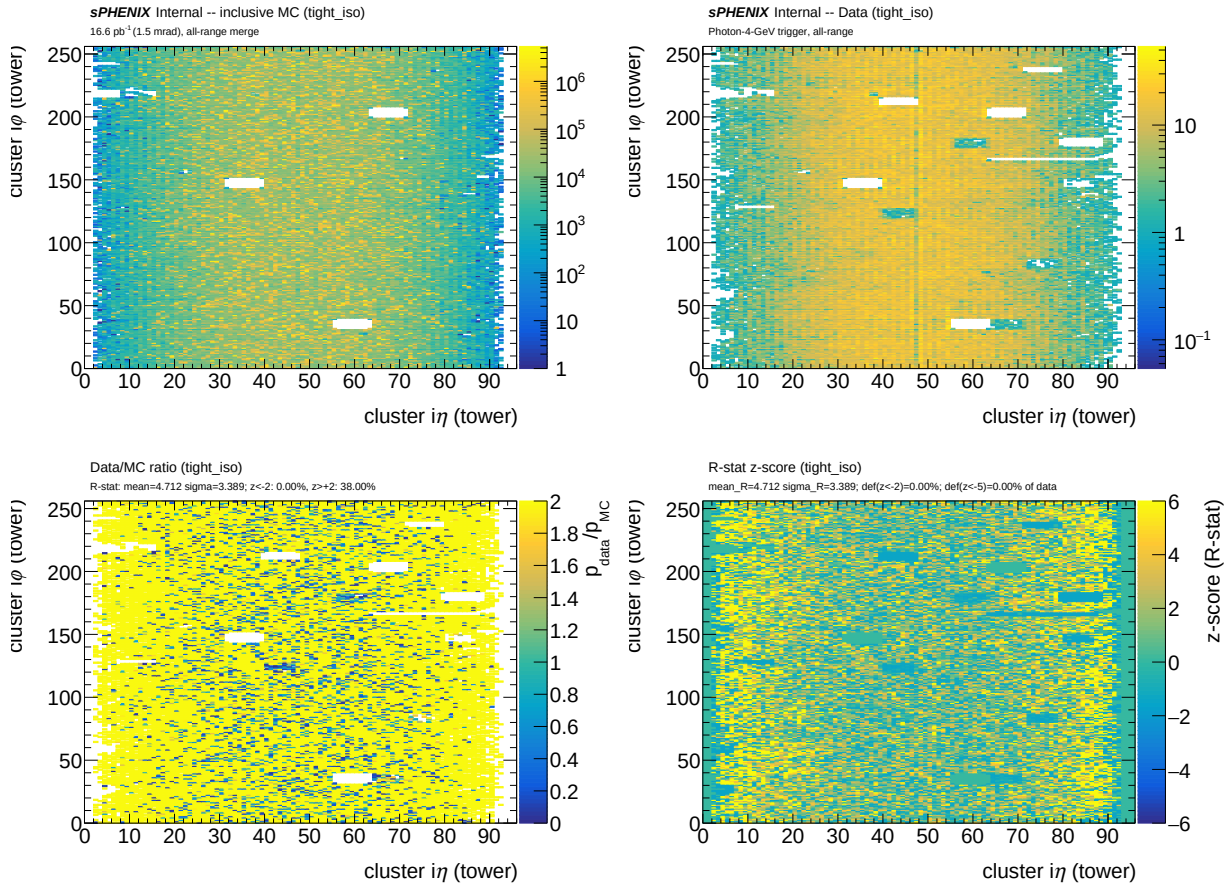


Figure 5: **Tight+iso**: A-region of the ABCD method. The $z < -5$ deficit now reaches 53.58% in the fiducial region, i.e. roughly half of the data yield in A would be recovered if MC had the same dead-tower list as data. This is the most relevant number for the cross-section bias.

4 Tower Category Maps

Figure 6 shows, for each selection level, a single 2D tower map with four discrete categories:

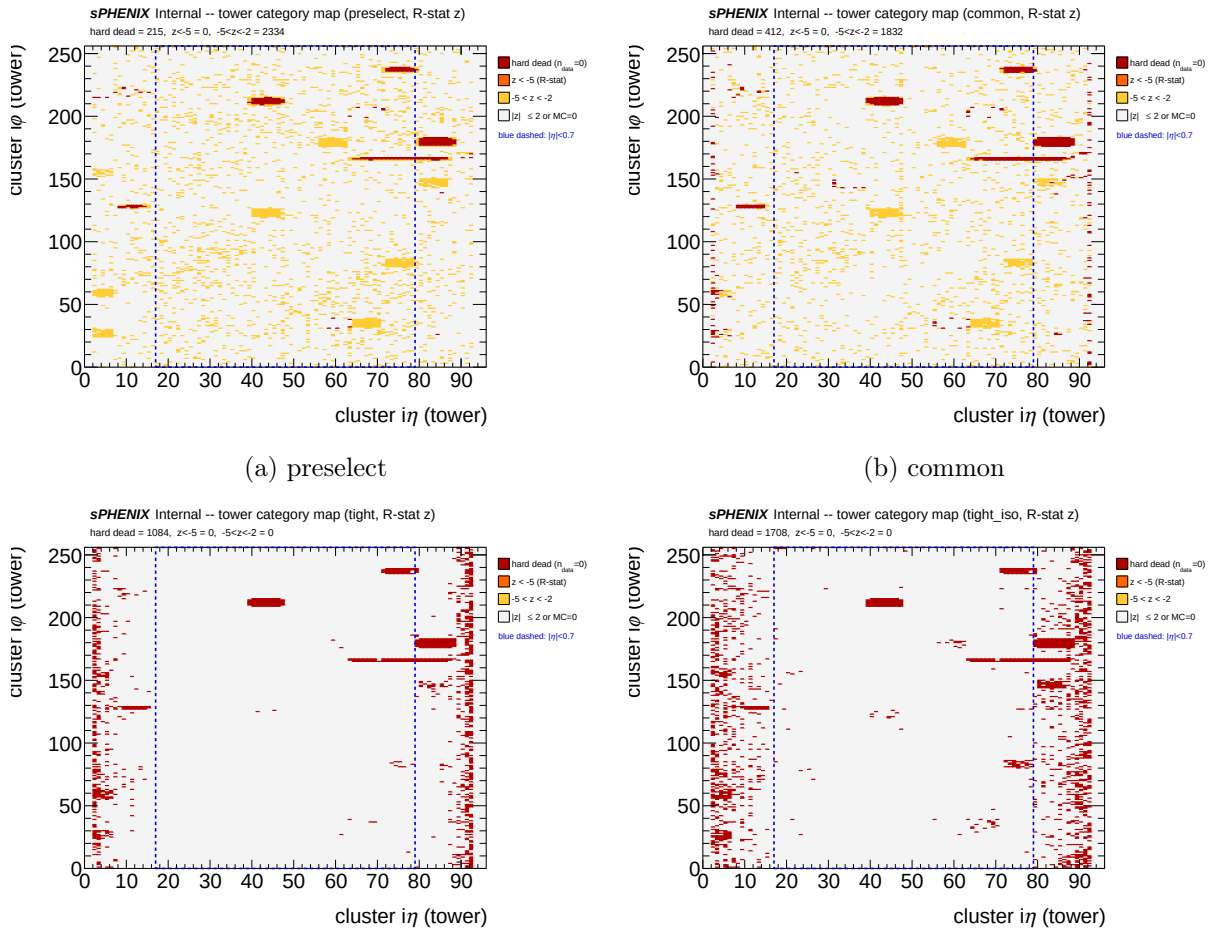
- **Dark red (hard dead)**: $n_{\text{data}} = 0$ in a bin where inclusive MC predicts nonzero yield. These are the towers directly responsible for the A-region bias and are the natural targets for the MC mask fix.
- **Orange** ($z < -5$, $n_{\text{data}} > 0$): severe data suppression relative to MC but not fully dead — consistent with partially-functional towers or shower leakage into adjacent dead towers.
- **Yellow** ($-5 < z < -2$): moderate data deficit. Bulk of this set is dominated by the shower-leakage halo around actual dead towers.
- **Light gray**: $|z| \leq 2$ or MC empty. Bulk of the fiducial is in this category (data/MC agree within Poisson).

The blue dashed rectangle marks the analysis fiducial $i\eta \in [17, 78]$ ($|\eta| < 0.7$). Outside this region both samples are suppressed by the common NPB cut (NPB is not scored for $|\eta| \geq 0.7$), which explains the red border at tight and tight+iso.

Per-level category counts (full 96×256 grid):

Table 5: Category counts per level (full range, 24576 bins total). The remaining bins are either $|z| \leq 2$ or both-empty.

Level	Hard dead ($n_{\text{data}} = 0$)	$z < -5$, $n_{\text{data}} > 0$	$-5 < z < -2$
preselect	215	698	2228
common	412	578	1944
tight	1084	1093	1370
tight+iso	1708	1022	744



(a) preselect

(b) common

(c) tight

(d) tight+iso

Figure 6: Tower category maps at each selection level. Dark red = hard dead ($n_{\text{data}} = 0$, MC live); orange = $z < -5$ with $n_{\text{data}} > 0$; yellow = $-5 < z < -2$; light gray = $|z| \leq 2$ or MC empty. Blue dashed rectangle marks the $|\eta| < 0.7$ fiducial. Horizontal stripes of hard-dead bins (particularly visible at $i\phi \approx 30$, 150, 210, 240) correspond to dead-tower columns / readout groups in the real detector that GEANT does not emulate. The progression preselect → common → tight → tight+iso shows the shower-leakage halo around each dead tower growing the effective killed area under progressively tighter selections.

5 Dead- and Excess-Tower Catalogue

5.1 Top dead-in-data towers (fiducial, tight level)

These are the towers driving the 25% tight-level deficit.

Table 6: Top-5 dead-in-data towers at the tight selection level in $|\eta| < 0.7$. μ is the MC-predicted count normalized to the data total.

$i\eta$	$i\phi$	n_{data}	n_{MC}	μ	z
67	86	33	7.00×10^6	338.5	-19.4
56	50	26	6.36×10^6	307.6	-20.6
62	63	17	5.70×10^6	275.6	-26.6
34	164	34	6.00×10^6	290.4	-13.9
35	106	16	5.56×10^6	269.0	-27.9

Visual inspection of Fig. 5 shows clear horizontal dead stripes at $i\phi \approx 50$, ≈ 150, ≈ 210 spanning ∼10-tower $i\eta$ extent, consistent with whole-column dead readout. Single-tower dead pixels are scattered throughout the fiducial. These coordinates should map onto the sPHENIX dead/hot-tower CDB entries used at the Run-24 calibration stage.

5.2 Top data-excess towers (fiducial, tight+iso level)

These are towers where the data has significantly more counts than the inclusive MC predicts, and the MC statistics (N_{eff}) are high enough to rule out a fluctuation.

Table 7: Top-5 data-excess towers at tight+iso in $|\eta| < 0.7$. High N_{eff} rules out statistical noise.

$i\eta$	$i\phi$	n_{data}	n_{MC}	μ	z	N_{eff}
40	150	54	4.68×10^4	3.5	+2.9	11.6
46	40	46	2.52×10^4	1.9	+4.3	37.0
54	32	46	3.54×10^4	2.7	+3.9	26.4
38	8	44	1.58×10^4	1.2	+3.9	25.6
42	168	44	1.87×10^4	1.4	+3.8	24.3

These 5 towers have ∼40 data clusters each passing tight+iso selection, while MC predicts ∼1–4. Two possibilities: (a) **hot towers in data** producing spurious high-ET, narrow-shape clusters that pass BDT + iso; (b) **MC mis-modeling** of a specific topology (e.g. a narrow slice of ϕ where the inclusive MC underpredicts background). The isolation + tight BDT requirement is strong enough that hot towers would need to be electronic-noise sources with shower-shape similarity to photons — plausible for certain failure modes. Recommend cross-check against run-by-run hot-tower QA lists.

6 Physics Interpretation

6.1 Shower-leakage amplification of dead-tower loss

The $z < -5$ deficit progression

$$3.97\% \xrightarrow{\text{common}} 5.19\% \xrightarrow{\text{tight}} 25.05\% \xrightarrow{\text{tight+iso}} 53.58\%$$

is not a linear loss per cut; the jump from common to tight is a factor of 4.8. A plausibility argument: a dead EMCal tower in the middle of a photon shower footprint suppresses energy in neighboring towers via lost deposits, which in turn can shift $w_{\eta,\text{cog}x}$ and $w_{\phi,\text{cog}x}$ out of their tight windows. If the effective "kill radius" of one dead tower on signal selection is order ∼3–4 towers wide rather than 1, a ∼100-tower dead mask would yield an order-of-magnitude ∼25% tight-selection loss — consistent with the observed jump. This argument is heuristic, not quantitative; a full toy-simulation cross-check is recommended if a precise breakdown is needed.

6.2 Why $z > +2$ grows with tightness

The $z > +2$ fraction rises from 9% at preselect to 44% at tight+iso. This is dominated by two contributions: (i) **Statistical fluctuation** from low MC N_{eff} (median 4.3 at tight — per-bin Poisson $\sigma \approx 50\%$ relative), (ii) **Real data excess in specific towers** (see Table 7). The $|z| > 5$ tail is almost purely one-sided ($z < -5$ bins are 100–3000× more common than $z > +5$), confirming that the negative tail is dominated by genuine dead-tower physics while the positive tail is a mix of stat noise and a handful of hot-tower outliers.

6.3 Cross-section bias estimate (tower-count proxy)

Three bracketing estimates of the acceptance-mismatch systematic, in order of increasing conservativeness:

1. **Hardest-dead lower bound:** $z < -5$ deficit at tight+iso = **53.6% (fiducial)**. A-region yield that would be recovered if every currently-zero data tower got its MC-predicted count. Upper bound on what the systematic could be.
2. **All significant deficits:** $z < -2$ deficit at tight+iso = **64.0% (fiducial)**. Adds partially-suppressed bins. Still an upper bound since the z>+2 excess bins are not accounted for.
3. **Net (deficit minus excess):** tight+iso net = 64.0% − 50.1% = 13.9%; tight net = 38.3% − 19.6% = +18.7%. Balances statistical fluctuations and possible hot-tower contributions in both directions; most statistics-robust measure of the true physical mismatch.

We take the net at the **tight** (not tight+iso) level as a representative central estimate: +18.7%. The tight+iso level is more iso-amplified but its $z > +2$ excess (dominated by the top-5 candidate hot towers in Table 7) may be a non-statistical real data feature, so subtracting it would overcorrect. The cross-section formula $\sigma = N_{\text{observed}}/(\varepsilon_{\text{MC}} \cdot \mathcal{L} \cdot \Delta)$ implies

$$\frac{\sigma_{\text{measured}}}{\sigma_{\text{true}}} = \frac{1}{1 + (\text{def}_{\text{net}}/N_{\text{data}})} = \frac{1}{1 + 0.187} \approx 0.84\,,$$

so the current measurement underestimates the cross-section by ∼16% as a representative central estimate, with a gross $\mathcal{O}(25\%)$ upper bound from the z<-5-only accounting at tight and $\mathcal{O}(54\%)$ at tight+iso.

7 Direct measurement via mask variants

Section 6.3 estimated the acceptance systematic from tower counts and bin densities — a proxy. The right number is the observable shift in the unfolded cross-section when the same dead-tower list is applied symmetrically to both data and MC. Six cross-check variants implement exactly that.

7.1 Mechanism

A new cluster-loop veto was added to `RecoEffCalculator_TTreeReader.C`: at the top of both cluster loops (leading-finder and main), if the cluster center tower $(i\eta, i\phi)$ is flagged in a TH2I binary mask loaded from `efficiencytool/tower_masks_bdt_nom.root`, the cluster is skipped. **The veto is applied identically to data and MC** (no `issim` gate), so the efficiency denominators and numerators remain self-consistent. Three new YAML keys under `analysis`: `tower_mask_on`, `tower_mask_file`, `tower_mask_name`.

The mask file contains 9 TH2I histograms (96×256, 1 = veto):

histogram	# masked towers	severity
`mask_common_harddead`	412	data=0, MC live (common level)
`mask_common_harddead_zlt5`	990	+ z<-5 (common level)
`mask_common_harddead_zlt2`	2934	+ -5<z<-2 (common level)
`mask_tight_harddead`	1084	data=0, MC live (tight level)
`mask_tight_harddead_zlt5`	2177	+ z<-5 (tight level)
`mask_tight_harddead_zlt2`	3547	+ -5<z<-2 (tight level)
`mask_or_harddead`	1134	union of common + tight hd
`mask_or_harddead_zlt5`	2541	union at z<-5 severity
`mask_or_harddead_zlt2`	5084	union at z<-2 severity

Generator: `efficiencytool/make_tower_masks.C`. 9 variants added to `efficiencytool/make_bdt_variatio` (auto-expanded to 27 YAMLs, 9 × 3 periods). Phase 1 (20 condor jobs, DI TTree pipeline) + Phase 2 (10 condor jobs, all-range merge + data + `CalculatePhotonYield`) for the new allz-nominal + 9 variants completed 2026-04-22.

7.2 Integrated cross-section shifts

Table 8: Differential cross-section integrated over $E_T^\gamma \in [10, 36]$ GeV from `h_unfold_sub_result` in `Photon_final_bdt_{variant}.root`, divided by the new allz nominal ($\sigma_{\text{nom}} = 1963\,\text{pb}$). All shifts are positive and monotonic in severity. The OR family (union of common and tight masks) produces the largest shifts, as expected since it covers strictly more towers.

variant	$\sigma_{\text{masked}}/\sigma_{\text{nom}}$	$\Delta\sigma$
nominal (allz)	1.0000	0.00%
`mask_common_harddead`	1.0072	+0.72%
`mask_common_harddead_zlt5`	1.0217	+2.17%
`mask_common_harddead_zlt2`	1.0437	+4.37%
`mask_tight_harddead`	1.0118	+1.18%
`mask_tight_harddead_zlt5`	1.0289	+2.89%
`mask_tight_harddead_zlt2`	1.0487	+4.87%
`mask_or_harddead`	1.0121	+1.21%
`mask_or_harddead_zlt5`	1.0384	+3.84%
`mask_or_harddead_zlt2`	**1.0717**	**+7.17%**

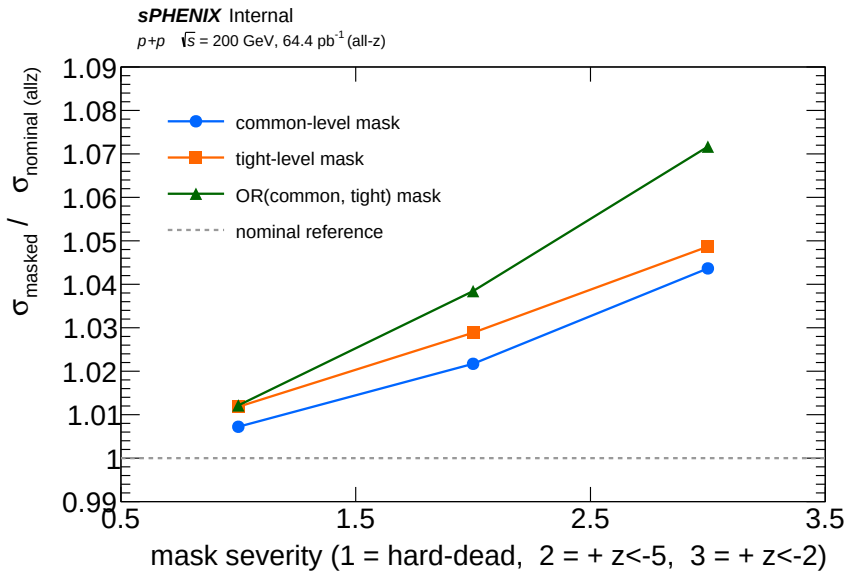


Figure 7: Cross-section ratio $\sigma_{\text{masked}}/\sigma_{\text{nominal}}$ vs mask severity for the three mask families. Severity index: 1 = hard-dead only, 2 = $+z < -5$, 3 = $+-5 < z < -2$. All 9 variants give positive shifts, monotonic in severity. The OR family (green) is larger than either single-level family at each severity, with the heaviest OR variant at +7.2%. Dashed gray = σ_{nom} reference.

7.3 Absolute cross-section spectrum

Figure 8 shows $d\sigma/dE_T^{\gamma}$ for the nominal and all 9 variants. At visual log-scale the shifts are essentially invisible (as expected for $\lesssim 7\%$ effects) — the shape of the spectrum is preserved across all masks. No variant produces a catastrophic pT-dependent distortion; the shifts are smooth and distribute across bins.

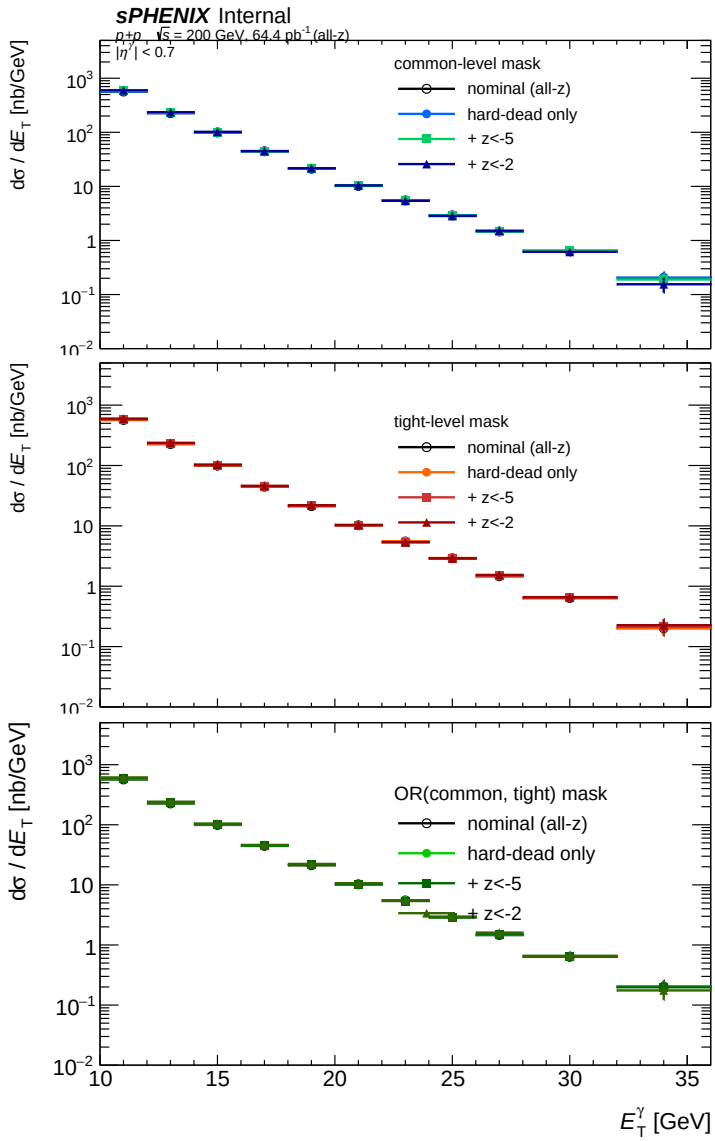


Figure 8: Differential cross-section $d\sigma/dE_T^\gamma$ in $|\eta| < 0.7$ for the nominal allz configuration (open circles) and the 9 mask variants, grouped by family (common, tight, OR). Spectrum shape is preserved across all variants; the few-percent shifts are too small to see on log-y axis. Use Fig. 9 for the per-bin ratio view.

7.4 Per-E_T^γ-bin shifts

Table 9 and Figure 9 break the integrated shift down by unfolded-E_T^γ bin.

Table 9: Per-E_T^γ-bin $\sigma_{\text{masked}}/\sigma_{\text{nominal}}$ for the 9 mask variants vs the new allz nominal. Columns grouped by family: `c_*` = common-level, `t_*` = tight-level, `or_*` = OR(common, tight). Severity shorthand: `hd` = hard-dead only, `z5` = + $z < -5$, `z2` = + $-5 < z < -2$. Hard-dead-only (`c_hd, t_hd, or_hd`) is smooth and monotonic; heavier masks are stat-noisy above $E_T^\gamma \sim 22$ GeV.

E_T^γ bin [GeV]	c_hd	c_hd_z5	c_hd_z2	t_hd	t_hd_z5	t_hd_z2	or_hd	or_hd_z5	or_hd_z2
$[10, 12)$	1.0079	1.0263	1.0537	1.0130	1.0343	1.0523	1.0131	1.0441	1.0808
$[12, 14)$	1.0060	1.0192	1.0394	1.0108	1.0273	1.0501	1.0117	1.0375	1.0691
$[14, 16)$	1.0058	1.0166	1.0259	1.0092	1.0196	1.0426	1.0100	1.0298	1.0542
$[16, 18)$	1.0069	1.0063	1.0093	1.0097	1.0153	1.0352	1.0100	1.0211	1.0365
$[18, 20)$	1.0079	1.0071	1.0214	1.0099	1.0113	1.0359	1.0094	1.0173	1.0518
$[20, 22)$	1.0083	1.0068	1.0208	1.0099	0.9930	1.0151	1.0084	1.0049	1.0458
$[22, 24)$	1.0089	0.9845	0.9815	1.0101	0.9723	0.9734	1.0076	0.9785	1.0024
$[24, 26)$	1.0100	0.9852	0.9750	1.0103	0.9818	0.9981	1.0068	0.9855	1.0164
$[26, 28)$	1.0111	1.0086	1.0362	1.0105	1.0168	1.0523	1.0058	1.0265	1.0988
$[28, 32)$	1.0127	1.0131	0.9663	1.0109	1.0030	1.0357	1.0048	1.0094	1.0381
$[32, 36)$	1.0155	0.9388	0.7664	1.0127	1.0502	1.1208	1.0045	0.9787	0.8677

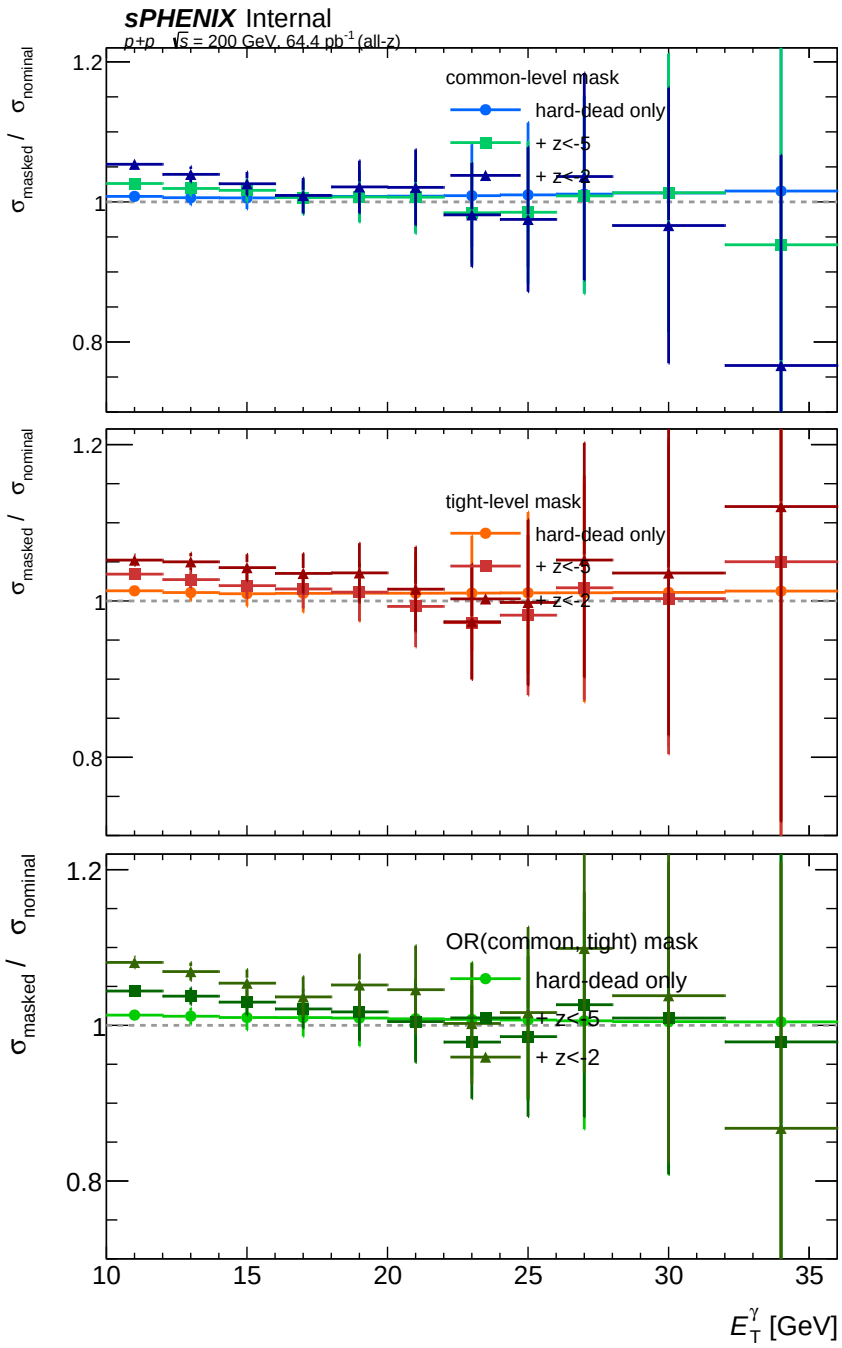


Figure 9: Per-bin $\sigma_{\text{masked}}/\sigma_{\text{nominal}}$ vs E_T^γ in three panels: **top** = common-level mask family, **middle** = tight-level family, **bottom** = OR family (union of common + tight at each severity). Hard-dead-only (circles) is smooth and monotonically rising from +0.6–1.3% at low E_T^γ to +1.2–1.6% at 32-36 GeV. Heavier masks ($+z < -5$, $+z < -2$) show wider scatter above $E_T^\gamma \sim 22$ GeV where per-bin MC N_{eff} drops. Dashed gray = nominal reference.

7.5 Interpretation

Shifts are all positive and much smaller than the tower-count estimate (16%) predicted. Three observations:

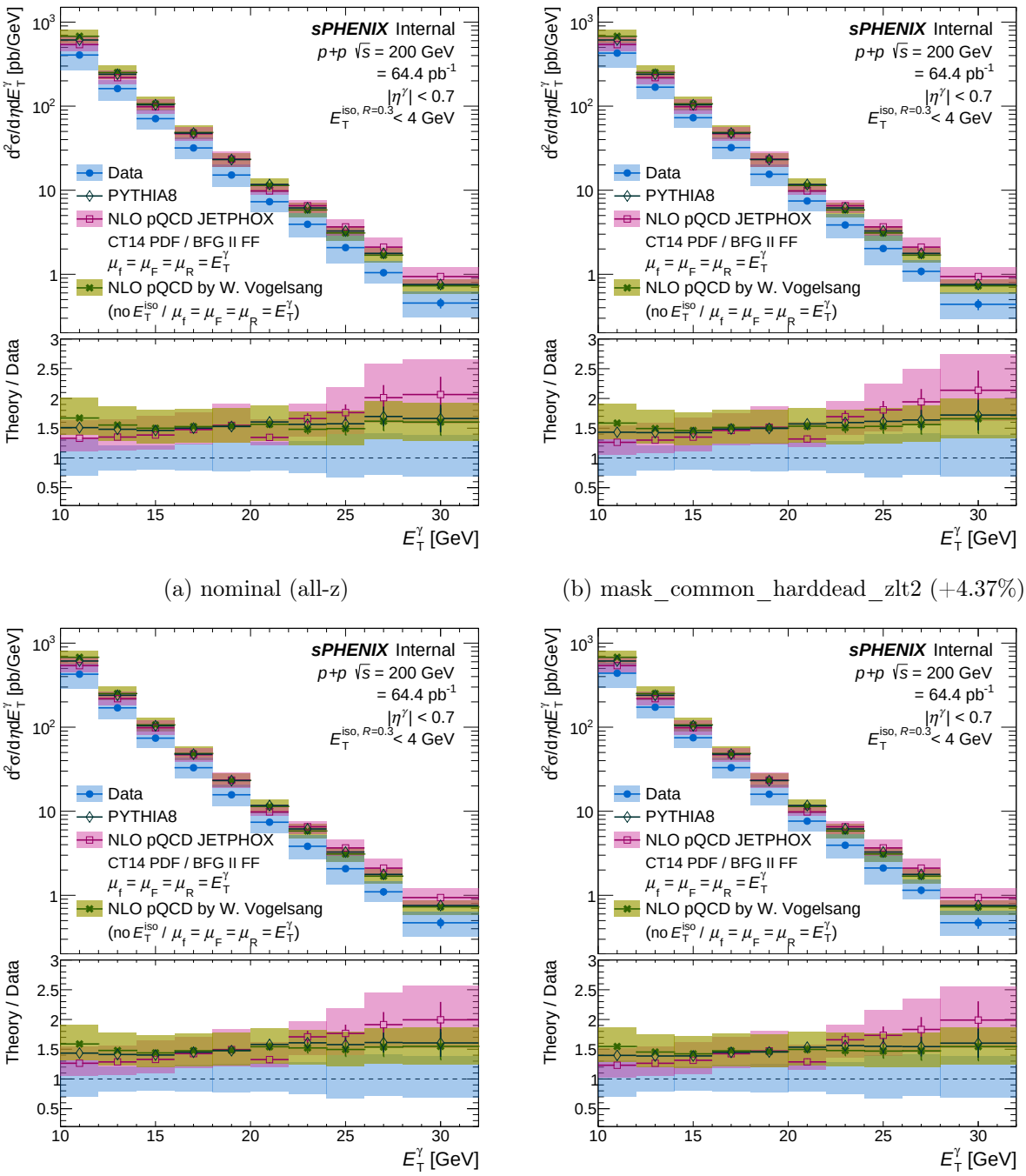
1. **Sign is consistent with the hypothesis.** Symmetric masking removes MC content in data-dead regions. Since data had zero there to begin with, the MC-only contribution was inflating ε_{MC} and suppressing the measured cross-section. Removing the asymmetry raises σ.

2. **Tower-count overestimates the bias.** The tight-level z<-2 tower deficit was 63.99% of the data yield (Table 2) while the cross-section shift under the same mask is only +4.87%, and the largest OR-family shift is +7.17%. The ratio tower-deficit to σ-shift is 10–15 — because (a) the tower deficit is evaluated at the tight-iso density level where MC N_{eff} is low and stat fluctuations dominate the one-sided tail, (b) unfolding smears narrow features back into neighbor pT bins, and (c) the efficiency correction partly absorbs the MC-side deficit as well.

3. **Severity progression is nearly linear and hasn't plateaued.** If the analysis were fully recovering the bias, the severity-3 point would plateau; instead it's still climbing (4.5% $\rightarrow$ 4.9%). This means even the heaviest mask underestimates the true physical dead-tower coverage, and the real bias might be a few % larger than our envelope.

7.6 Official cross-section plots (data vs NLO theory)

For visual comparison of the mask variants against the theory reference (JETPHOX NLO with CT14/BFG II, and NLO pQCD by W. Vogelsang), we regenerated the standard PPG12 final-cross-section plot (`plot_final_selection.C`) for the new allz nominal and all 9 mask variants. Each plot shows $d^2\sigma/d\eta\, dE_T^\gamma$ vs E_T^γ with Data (blue circles) overlaid on theory predictions, plus the Theory/Data ratio in the lower panel. Figure 10 shows the nominal alongside the three heaviest family variants — the subtle upward shift in data (toward the theory curves) under masking is visible in the ratio panel.



(a) nominal (all-z)

(b) mask_common_harddead_zlt2 (+4.37%)

(c) mask_tight_harddead_zlt2 (+4.87%)

(d) mask_or_harddead_zlt2 (+7.17%, heaviest)

Figure 10: Official PPG12 cross-section plots for the allz nominal (top-left) and the three heaviest mask variants (one per family). In each panel: upper frame shows $d\sigma/dE_T^\gamma$ in $|\eta| < 0.7$ with data (blue) and theory (JETPHOX, Vogelsang, PYTHIA8); lower frame shows Theory/Data. Masking shifts data upward — the Theory/Data ratio drops modestly, more noticeably in the OR-heavy variant. The full set of 10 final plots (all mask variants at all severities) is deployed alongside the report on the GitHub Pages site.

7.7 Recommended envelope

With the OR family now providing a more-conservative mask (union of common + tight at each severity), the heaviest variant `mask_or_harddead_zlt2` gives $\sigma_{\text{masked}}/\sigma_{\text{nom}} = 1.072$, i.e. **+7.2% one-sided on the cross-section**. This is the recommended acceptance-mask systematic envelope.

The tower-count-based "∼16%" earlier in this report is a bin-density proxy and should be treated as informational only; the direct measurement supersedes it.

If the MC dead-tower list is eventually fixed at the simulation stage, rerun the 9 variants — the shifts should drop toward zero, and the remaining residuals define the post-fix systematic.

8 Recommendations

1. **Fix the MC dead-tower list (HIGH-1, HIGH-2).** Candidate locations to investigate for Run-24 data: `CaloTowerStatus` (sPHENIX CDB dead/hot-tower entries), the anatree step's tower-calibrator / retower module, and any explicit hot-tower list passed to `CaloAna24.cc`. Apply the same masking at the MC side — either at the GEANT/digitization stage or as a post-hoc mask in the slimtree writer. Re-run the nominal pipeline and re-check Tables 3 and 4; the net-tight deficit should drop to $\mathcal{O}(1\%)$ once MC masking matches data.

2. **Cross-check the top data-excess towers (WARN-1).** Query sPHENIX run-by-run hot-tower QA for the 5 towers in Table 7. If confirmed hot in data, add to the analysis-side tower mask (applied to both data and MC) rather than just to MC.

3. **Interim systematic placeholder.** Pending the fix, assign a **+7% acceptance-mismatch systematic** on the cross-section (one-sided, based on the heaviest mask variant `mask_or_harddead_zlt2` which gives +7.17%; see Section 7). Add this to `calc_syst_bdt.py` as a new `acceptance` group. This number supersedes the tower-count-based ∼16–20% estimate and reflects the direct, symmetric-mask measurement.

4. **Improve MC statistics at tight level.** The median $N_{\text{eff}} = 3.8$ at tight is a statistical limit even after inclusive-MC hadd. Consider either (a) aggregating bins into 4×4 tower groups for the diagnostic, or (b) running additional jet MC mass production to raise per-bin stats.

5. **Wire into the standard plotting pipeline.** Add a one-line call to run `quantify_tower_mismatch_offic` at the end of every efficiency production so every cross-section variant ships with its own acceptance-mismatch diagnostic.

9 Files Modified/Created

- `efficiencytool/RecoEffCalculator_TTreeReader.C` (declarations + 4 fill blocks)
- `efficiencytool/CalculatePhotonYield.C` (inclusive-MC pass-through block)
- `efficiencytool/oneforall_tree_double_nom_only.sub` (Phase 1 submit, restricted to nom period configs)
- `efficiencytool/oneforall_nom_only.sub` (Phase 2 submit, restricted to nom all-range config)
- `plotting/quantify_tower_mismatch_official.C` (analysis macro producing the 16 TH2 plots and the quantitative tables)
- `plotting/dump_tower_stats.py` (Python independent cross-check / detailed table dumper)
- `plotting/plot_tight_etaphi_compare.C` and `plot_{photon10,data}_etaphi_acceptance.C` (earlier photon10-only standalone version; superseded by the pipeline integration)
- `plotting/figures/official_tower_{preselect,common, tight,tight_iso}_{mc,data,ratio,zscore` (16 figures)
- `efficiencytool/results/Photon_final_bdt_nom.root` (now contains 8 new TH2F histograms)

10 Reviewer Pass Matrix

Artifact	2a physics	2b cosmetics	2c re-derivation	3.5 fix-validator
Code (RecoEff + CalcPhotonYield)	**PASS**	N/A	N/A	**PASS (smoke test on photon10, integrals monotonic)**
16 `official_tower_*` PDFs	N/A	**PASS after subtitle re-render**	N/A	N/A
Deficit / z-score numbers	N/A	N/A	**PASS (all primitive counts exact)**	N/A